

DropBox

Functional requirements :

user should be able to upload/download any file based on his preference,
user should be able to search for any file based on his preference,
user should be able to sync all the upload/download operations

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughout,
CAP theorem (Here Eventual consistency is fine , but here we need 99.99% uptime i.e
Availability>>Consistency)

Core Entities:

user,
file,
history

API :

POST:/upload

```
{  
  userID,  
  filemetadata  
}
```

GET:/search/{fileID}

HLD:

